

Nourish Database System - Project Reflection

Group work and collaboration

Our group worked together by dividing the project into three main areas: database design, application coding, and documentation/testing. One group member focused mainly on creating the SQLite schema and making sure the tables matched the project requirements. Another group member worked on the Python menu system and the functions for adding and viewing records. The third group member helped test the system, check that the features worked, and prepare the final project materials. We collaborated using GitHub to share the source code and kept each other updated through messages and in-person discussion during class work time.

Use of AI

We did use AI as a support tool while coding and preparing the project. We decided to use it because it helped us understand how to structure Python functions, write SQLite queries, and check whether the project matched the course requirements. AI was useful for explaining errors, suggesting cleaner code organization, and helping draft documentation. However, significant human work was still required. We had to decide what our system should actually do, design the food pantry database, choose the tables and relationships, test the program with our own data, and make sure the final version made sense for our project.

Integration issues

Our main integration challenge was connecting the command-line interface to the SQLite database in a way that was easy to use and did not break foreign key relationships. For example, visits had to connect to existing households and volunteers, and distribution items had to connect to existing inventory lots. We had to think carefully about the order users enter data so that records exist before they are referenced. We did make small adjustments while building, especially around inventory lots, visit history, proxy pickups, and calculating remaining inventory. The hardest part was making the database design feel like a real working system instead of just separate tables.